

EJERCICIO 3

Efectúa las siguientes integrales:

1. $\int x^6 dx$
2. $\int 5x^4 dx$
3. $\int bx^3 dx$
4. $\int \sqrt{3}x^2 dx$
5. $\int a dx$
6. $\int \frac{3 dx}{4}$
7. $\int \frac{dx}{3}$
8. $\int \sqrt[3]{x} dx$
9. $\int 5\sqrt[4]{x} dx$
10. $\int \frac{dx}{x^3}$
11. $\int \frac{5 dx}{x^4}$
12. $\int \frac{4 dx}{x}$
13. $\int \frac{dx}{\sqrt[4]{x}}$
14. $\int \frac{6 dx}{\sqrt[3]{x}}$
15. $\int \sqrt[5]{x^3} dx$
16. $\int \frac{a dx}{\sqrt[3]{x^2}}$
17. $\int \frac{5 dx}{2x}$
18. $\int \sqrt{bx} dx$
19. $\int \left(\frac{5}{\sqrt[3]{x}} - 4\sqrt[3]{x} \right) dx$
20. $\int \left(\frac{3}{x^5} - \frac{2}{x^2} - \frac{6}{x} \right) dx$
21. $\int \sqrt[3]{at} dt$
22. $\int \sqrt{6t} dt$
23. $\int (8x^5 - 5x^4 - 4x^3 - 6x^2 - 2x - 3) dx$
24. $\int (ax^3 - bx^2 - cx + d) dx$
25. $\int \left(\frac{x^2}{\sqrt{a^2 + b^2}} - \frac{3x}{\sqrt{a}} - 5\sqrt{b} \right) dx$
26. $\int \left(\frac{x^4 - 6x^3 - 7x}{x} \right) dx$
27. $\int \left(\frac{3}{\sqrt[5]{x^2}} - \frac{2}{\sqrt[5]{x}} \right) dx$
28. $\int \left(\frac{4}{\sqrt[3]{x}} - \frac{5}{\sqrt[4]{x}} \right) dx$
29. $\int \left(y^{\frac{5}{2}} - 5y^{\frac{4}{3}} - 2y^{\frac{1}{4}} - \sqrt{y} \right) dy$
30. $\int \left(\frac{y^{\frac{7}{2}} - y^{\frac{5}{3}} - y^{\frac{1}{4}}}{y^2} \right) dy$
31. $\int \sqrt[3]{t}(5t^2 - 3t + 2) dt$
32. $\int \sqrt[3]{7t} dt$
33. $\int (3x + 4)^6 dx$
34. $\int (ax^2 - b)^5 x dx$
35. $\int t^2(t^3 - 4)^2 dt$
36. $\int (a - by)^4 dy$
37. $\int (t^2 - 6)^2 dt$
38. $\int x(x + 4)^2 dx$

$$39. \int x^2(x+1)^3 dx$$

$$40. \int \sqrt{m+ny} dy$$

$$41. \int \sqrt{5x-3} dx$$

$$42. \int \frac{t dt}{\sqrt{at^2+b}}$$

$$43. \int \frac{dx}{\sqrt[3]{9x-1}}$$

$$44. \int (\sqrt{x}-4)^2 dx$$

$$45. \int \frac{x dx}{(3x^2-4)^4}$$

$$46. \int \frac{5 dx}{(3x-4)^2}$$

$$47. \int \frac{8x dx}{(2x^2+5)^4}$$

$$48. \int \frac{(\sqrt{x}-b)^2}{\sqrt{x}} dx$$

$$49. \int \frac{dt}{at+b}$$

$$50. \int \frac{x dx}{3x^2-4}$$

$$51. \int \frac{dx}{x+3}$$

$$52. \int \frac{4x dx}{2x^2-6}$$

$$53. \int \frac{(2x-3) dx}{(x^2-3x+6)^2}$$

$$54. \int (x^2-2)\sqrt{x^3-6x+3} dx$$

$$55. \int \frac{y^{n-1} dy}{(ay^n+b)^m}$$

$$56. \int e^{3x}(1-e^{3x})^2 dx$$

$$57. \int \frac{(4-\ln|x+3|)^3 dx}{x+3}$$

$$58. \int \cos 4x(1-\sin 4x)^3 dx$$

$$59. \int \csc^2 x \sqrt{3+\cot x} dx$$

$$60. \int \frac{\sec 2x \tan 2x}{\sqrt{1-\sec 2x}} dx$$

$$61. \int \frac{\cos ax}{1-\sin ax} dx$$

$$62. \int \frac{e^{\sqrt{x}} \sqrt{e^{\sqrt{x}}}-1}{\sqrt{x}} dx$$

$$63. \int \cot x(2+\ln|\sin x|) dx$$

$$64. \int \frac{\sin 2x dx}{(1-\cos^2 x)^3}$$

$$65. \int \sin^2 bx \cos bx dx$$

$$66. \int \cot mx \csc^2 mx dx$$

$$67. \int \cos^2 4x \sin 4x dx$$

$$68. \int \frac{\cos 5x}{\sqrt{\sin 5x+4}} dx$$

$$69. \int \frac{4x+2}{x+2} dx$$

$$70. \int \frac{(3x^2+2) dx}{x-1}$$

$$71. \int \frac{dy}{y \ln^2 y}$$

$$72. \int \frac{dx}{2x \ln 3x}$$

$$73. \int x^n \sqrt{ax^{n+1}+b} dx$$

$$74. \int \frac{1}{x^3} \sqrt{1-\frac{1}{x^2}} dx$$

$$75. \int \csc^2 3x \cos 3x dx$$

$$76. \int \left(\frac{2}{(x+1)^3} - \frac{3}{(x+1)^2} + \frac{4}{x+1} \right) dx$$

$$77. \int \left(\frac{3}{x+2} - \frac{4}{x+5} \right) dx$$

$$78. \int \left(\frac{3}{2x-1} + \frac{5}{3x-4} \right) dx$$

$$79. \int \frac{\operatorname{sen} x}{\sqrt[3]{\cos^2 x}} dx$$

$$80. \int \operatorname{sen}^3 x \operatorname{sen} 2x dx$$

$$81. \int \frac{dw}{\operatorname{sen}^2 w \sqrt{1 - \cot w}}$$

$$82. \int \frac{3 \operatorname{sen} y \cos y}{\sqrt{1 - 2 \operatorname{sen}^2 y}} dy$$

$$83. \int \sqrt{1 + \cos \alpha} d\alpha$$

$$84. \int \frac{\operatorname{sen}^{\frac{3}{4}} x}{\cos^{\frac{11}{4}} x} dx$$